

|          |            |           |             |            |           |
|----------|------------|-----------|-------------|------------|-----------|
| AMPACITY | MVA RATING | COND TEMP | SHEATH TEMP | TOTAL LOSS | STATUS    |
| 993 A    | 227.10 MVA | 86.6 degC | 79.1 degC   | 37.983 W/m | Converged |

1. Design Data & Input Parameters

| Parameter                         | Formula / Symbol | Value    | Unit  |
|-----------------------------------|------------------|----------|-------|
| Conductor material                |                  | Cu       |       |
| Conductor size                    |                  | 1200     | mm2   |
| Conductor type                    |                  | milliken |       |
| Conductor diameter Dc             |                  | 43.1     | mm    |
| Conductor screen OD Dc_sc         |                  | 46.7     | mm    |
| Max conductor temp theta_max      |                  | 90       | degC  |
| Insulation material               |                  | XLPE     |       |
| Insulation screen OD Di           |                  | 80.5     | mm    |
| Ins outer screen OD Di_sc         |                  | 82.9     | mm    |
| Insulation thickness t_ins        |                  | 16.10    | mm    |
| Relative permittivity er          |                  | 2.5000   |       |
| Loss tangent tan(delta)           |                  | 0.001000 |       |
| Screen type                       |                  | Cu_wire  |       |
| Screen mean diameter ds           |                  | 84.3     | mm    |
| Screen area As                    |                  | 140.2    | mm2   |
| Armour type                       |                  | none     |       |
| Oversheath material               |                  | PE       |       |
| Cable outer diameter De           |                  | 98.0     | mm    |
| Rated voltage U                   |                  | 132.0    | kV    |
| U0 line-to-earth                  |                  | 76.210   | kV    |
| Frequency                         |                  | 50       | Hz    |
| Bonding                           |                  | solid    |       |
| Formation                         |                  | trefoil  |       |
| Depth to cable centre L           |                  | 1000     | mm    |
| Axial spacing s                   |                  | 98.0     | mm    |
| Ambient ground temp theta_a       |                  | 20       | degC  |
| Soil thermal resistivity rho_soil |                  | 1.00     | K.m/W |

2. DC Resistance [IEC 60287-1-1 Cl.2.1.1]

| Parameter                   | Formula / Symbol | Value      | Unit  |
|-----------------------------|------------------|------------|-------|
| IEC 60228 R0 at 20 degC     |                  | 1.5100e-05 | ohm/m |
| Temperature coefficient a20 |                  | 3.9300e-03 | 1/K   |
| R0 at theta_amb = 20 degC   |                  | 1.5100e-05 | ohm/m |
| R0 at theta_max = 90 degC   |                  | 1.9254e-05 | ohm/m |

3. AC Resistance at theta\_max [IEC 60287-1-1 Cl.2.1.2, 2.1.4]

| Parameter                                 | Formula / Symbol | Value      | Unit  |
|-------------------------------------------|------------------|------------|-------|
| ks strand constant                        |                  | 0.4350     |       |
| kp proximity constant                     |                  | 0.3700     |       |
| $xs4 = [ks(8\pi if/R0) \times 10^{-7}]^2$ |                  | 8.060386   |       |
| ys skin effect factor                     |                  | 0.000000   |       |
| $xp4 = [kp(8\pi if/R0) \times 10^{-7}]^2$ |                  | 5.831507   |       |
| yp" intermediate proximity                |                  | 0.029652   |       |
| yp proximity factor                       |                  | 0.022931   |       |
| R at theta_max AC = $R0(1+ys+yp)$         |                  | 1.9696e-05 | ohm/m |

#### 4. Dielectric Losses [IEC 60287-1-1 Cl.2.2]

| Parameter                                                     | Formula / Symbol | Value        | Unit  |
|---------------------------------------------------------------|------------------|--------------|-------|
| Relative permittivity er                                      |                  | 2.5000       |       |
| Loss tangent tan(delta)                                       |                  | 0.001000     |       |
| U0 line-to-earth                                              |                  | 76.2100      | kV    |
| Capacitance $C = er / (18 \ln(Di\_sc/Dc\_sc)) \times 10^{-9}$ |                  | 0.242013     | nF/m  |
| $\omega = 2 \pi f$                                            |                  | 314.1593     | rad/s |
| $Wd = \omega C U0^2 \tan(\delta)$                             |                  | 4.415826e-01 | W/m   |
| TB880 GP7                                                     |                  | Applied      |       |

#### 5. Screen / Sheath Setup [IEC 60287-1-1 Cl.2.3]

| Parameter                        | Formula / Symbol | Value      | Unit  |
|----------------------------------|------------------|------------|-------|
| Screen material                  |                  | Cu         |       |
| Screen resistivity rho_s         |                  | 1.7241e-08 | ohm.m |
| Screen area As                   |                  | 140.25     | mm2   |
| Lay factor LFs                   |                  | 1.136863   |       |
| Screen Rs0 at 20 degC (with LFs) |                  | 1.3976e-04 | ohm/m |
| --- Al laminated foil ---        |                  |            |       |
| Foil thickness t_fl              |                  | 0.200      | mm    |
| Foil outer diameter D_fl         |                  | 88.8       | mm    |
| Foil area A_fl                   |                  | 55.6690    | mm2   |
| Foil Rso_fl at 20 degC           |                  | 5.0772e-04 | ohm/m |

#### 6. Armour Losses [IEC 60287-1-1 Cl.2.4]

| Parameter   | Formula / Symbol | Value    | Unit |
|-------------|------------------|----------|------|
| Armour type |                  | None     |      |
| lam2        |                  | 0.000000 |      |

#### 7. Thermal Resistances [IEC 60287-2-1]

| Parameter                                    | Formula / Symbol | Value        | Unit  |
|----------------------------------------------|------------------|--------------|-------|
| rho_ins insulation                           |                  | 3.50         | K.m/W |
| rho_sc semiconducting screens                |                  | 2.50         | K.m/W |
| rho_tape bedding tape under wire screen      |                  | 12.00        | K.m/W |
| T1_cs conductor screen                       |                  | 0.0319189884 | K.m/W |
| T1_i insulation                              |                  | 0.3033167853 | K.m/W |
| T1_si insulation screen                      |                  | 0.0116890861 | K.m/W |
| T1_uwbt bedding tape under wire screen       |                  | 0.0319840375 | K.m/W |
| T1_raw = sum of all layers                   |                  | 0.3789088974 | K.m/W |
| T1 correction x1.00 (foil=True, cover=42.4%) |                  | 0.3789088974 | K.m/W |
| T2 screen to foil tape (0 if no foil)        |                  | 0.0906994188 | K.m/W |
| T3_raw oversheath                            |                  | 0.0537757090 | K.m/W |
| T3 x1.6 touching trefoil factor              |                  | 0.0860411344 | K.m/W |
| u = 2L/De                                    |                  | 20.408163    |       |
| T4 soil                                      |                  | 1.4701534385 | K.m/W |
| Sum T = T1+T2+T3+T4                          |                  | 2.0258028891 | K.m/W |

## 8. Sheath Losses at Convergence [IEC 60287-1-1 Cl.2.3]

| Parameter                 | Formula / Symbol | Value      | Unit  |
|---------------------------|------------------|------------|-------|
| theta_s at convergence    |                  | 79.1083    | degC  |
| Rs at theta_s             |                  | 1.3519e-04 | ohm/m |
| X sheath reactance        |                  | 5.2038e-05 | ohm/m |
| lam1' circulating current |                  | 0.885796   |       |
| lam1" eddy current        |                  | 0.046085   |       |
| lam1 = lam1' + lam1"      |                  | 0.931881   |       |
| lam2 armour               |                  | 0.000000   |       |

## 9. Ampacity Calculation [IEC 60287-1-1 Cl.1.4]

| Parameter                           | Formula / Symbol | Value      | Unit |
|-------------------------------------|------------------|------------|------|
| Delta_theta = theta_max - theta_amb |                  | 70.00      | K    |
| Etop numerator                      |                  | 69.189100  |      |
| Ebot denominator                    |                  | 7.0126e-05 |      |
| I = sqrt(Etop/Ebot) exact           |                  | 993.2963   | A    |
| I rounded TB880 GP2                 |                  | 993        | A    |
| MVA = sqrt(3) x U x I               |                  | 227.098    | MVA  |
| Iterations                          |                  | 5          |      |
| Convergence                         |                  | YES        |      |

## 10. Temperature Profile Through Cable

| Parameter                   | Formula / Symbol | Value   | Unit |
|-----------------------------|------------------|---------|------|
| theta_c conductor target    |                  | 90.0    | degC |
| theta_c back-calculated     |                  | 86.5550 | degC |
| theta_s sheath surface      |                  | 79.1083 | degC |
| theta_j cable outer surface |                  | 75.8402 | degC |
| theta_a ambient ground      |                  | 20.0    | degC |

## 11. Loss Budget

| Parameter                         | Formula / Symbol | Value      | Unit |
|-----------------------------------|------------------|------------|------|
| Conductor Wc = I <sup>2</sup> x R |                  | 19.432345  | W/m  |
| Sheath Ws = lam1 x Wc             |                  | 18.108634  | W/m  |
| Armour Wa = lam2 x Wc             |                  | 0.000000   | W/m  |
| Dielectric Wd                     |                  | 4.4158e-01 | W/m  |
| Total                             |                  | 37.982561  | W/m  |

12. Electrical Parameters

| Parameter             | Formula / Symbol | Value    | Unit   |
|-----------------------|------------------|----------|--------|
| Inductance L          |                  | 0.168747 | uH/m   |
| Reactance X = omega L |                  | 53.0133  | uOhm/m |
| Pos-seq resistance R1 |                  | 38.0494  | uOhm/m |
| Pos-seq impedance Z1  |                  | 65.2546  | uOhm/m |
| Capacitance C         |                  | 0.242013 | nF/m   |
| Charging current Ic   |                  | 5.7943   | mA/m   |

Standards: IEC 60287-1-1:2023 | IEC 60287-2-1:2015 | CIGRE TB 880 guidance applied.  
GP2: ampacity rounded down. GP6: eddy losses always calculated. GP7: dielectric losses always calculated. GP8: exact T4 formula. T3: 1.6x factor for touching trefoil.  
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